

Governed Clinical Analytics Copilot

OmniSQL-7B Fine-Tuning on EHRSQL 2024 MIMIC-IV

Project Report · July 2026

RS(10)	EX	Abstentions	Hallucinations	vs Competition Winner
0.873	92.4%	226 / 233 (97.0%)	7 / 233 (3.0%)	+6.0 pp (0.813 → 0.873)

Test set: EHRSQL 2024 MIMIC-IV official split — 1,167 questions (934 answerable + 233 unanswerable). RS(10) = (correct SQL + correct abstentions – 10 × hallucinations) / total. Model: OmniSQL-7B SFT (LoRA r=32, α=64, 2 epochs, 53 K examples). Zero-shot inference with full MIMIC-IV schema prompt.

1. Benchmark Comparison

All known RS(10) results on the EHRSQL 2024 MIMIC-IV official test split (1,167 questions). Post-competition literature search covered ~70 citing papers and all major NL2SQL papers through July 2026.

System	RS(10)	EX	Model / Approach	Source
This work (OmniSQL-7B SFT)	0.873	92.4%	OmniSQL-7B, LoRA SFT only	This project
LG AI Research + KAIST [1st]	0.813	88.2%	GPT-3.5-turbo SFT + self-training + entropy filter	arXiv 2405.11162
PromptMind [2nd]	0.749	—	GPT-3.5 + GPT-4 + Claude Opus ensemble	arXiv 2405.08839
ProbGate [3rd]	0.742	—	GPT-4 + log-prob threshold filtering	arXiv 2404.16659
KU-DMIS [4th]	0.592	—	T5-3B + question templatzation	arXiv 2406.00014
Qwen2.5-Coder-7B ORPO v4 + entropy filter	0.409	35.6%	Qwen2.5-Coder-7B ORPO + entropy threshold gate	This project
Qwen2.5-Coder-7B ORPO v4	0.128	40.5%	Qwen2.5-Coder-7B ORPO v4, no filter	This project
OmniSQL-7B zero-shot (pre-SFT baseline)	-0.564	61.2%	OmniSQL-7B, no fine-tuning, k=10 few-shot	This project

Note: CBR-to-SQL (arXiv 2603.05569, March 2026) achieves EX=89.5% on the same split but has no abstention mechanism — RS(10) is undefined. RaQAD (CIKM 2025) is paywalled and unconfirmed. The Codabench leaderboard requires login; post-deadline submissions may exist.

2. Model Progression

All experiments on the EHRSQL 2024 MIMIC-IV Demo test set — 1,167 questions (934 answerable + 233 unanswerable). Halluc. = SQL answers produced for unanswerable questions. Abstain = correct_abstentions / 233 unanswerable total.

Stage	EX	RS(0)	RS(5)	RS(10)	Halluc.	Abstain
Qwen2.5-Coder-7B ORPO v4 (baseline)	40.5%	48.8%	30.8%	0.128	42	191/233
Qwen2.5-Coder-7B ORPO v4 + entropy filter	35.6%	47.7%	44.3%	0.409	8	225/233
OmniSQL-7B zero-shot (pre-SFT baseline)	61.2%	57.6%	0.6%	-0.564	133	100/233
OmniSQL-7B SFT (final) ★	92.4%	—	—	0.873	7	226/233

★ Colab A100. Teal rule separates Qwen2.5-Coder-7B experiments from OmniSQL-7B experiments.

3. Retrieval System (Few-Shot Example Retrieval)

Few-shot retrieval is evaluated on the 934 answerable questions from the EHRSQL 2024 MIMIC-IV test set against a corpus of 40,030 train+aug examples. Relevance is defined at the 'medium' label level (same SQL structure / question type). Hit@k = at least one relevant example in the top-k results. MRR = Mean Reciprocal Rank of the first relevant hit.

Method	Hit@1	Hit@2	Hit@3	Hit@5	Hit@10	MRR
BM25	55.8%	59.2%	66.5%	70.1%	74.1%	61.2%
Dense Embed (BGE-Large)	68.5%	71.6%	75.4%	79.1%	81.0%	72.4%
Hybrid BM25 + Embed ★	68.8%	72.1%	75.5%	78.1%	81.6%	72.6%
Template Classifier (LogReg)	59.6%	59.6%	59.6%	59.6%	59.6%	59.6%

Reranking Study

Cross-encoder rerankers trained on web data (MS-MARCO, BGE) consistently degraded performance versus the hybrid baseline. PRF (Pseudo-Relevance Feedback) pivot reranking came closest but still did not beat the baseline. A task-specific trained reranker is needed to improve beyond 72.1% Hit@2.

Method	Hit@2	nDCG@2	MRR	vs Baseline
Hybrid baseline (no reranking)	72.1%	68.4%	70.5%	—
Hybrid + PRF top-1 pivot @20	69.2%	68.9%	69.0%	−2.9 pp
Hybrid + PRF top-3 vote @50	69.1%	69.0%	69.0%	−3.0 pp
Hybrid + MS-MARCO rerank @100	69.9%	66.3%	68.4%	−2.2 pp
Hybrid + MS-MARCO rerank @500	68.8%	65.3%	67.4%	−3.3 pp
Hybrid + BGE cross-encoder @20	68.3%	64.2%	66.1%	−3.8 pp

Recall Ceiling Analysis

Recall @ k	Hybrid	Interpretation
k = 2	72.1%	Usable for 2-shot prompting
k = 10	81.6%	Diminishing return over k=2
k = 100	86.6%	Max first-stage recall at k=100
k = 500	88.1%	Near corpus ceiling
k = 1000	88.9%	Diminishing; not practical
Label ceiling	89.8%	10.2% of queries have NO relevant example in the 40K corpus — uncoverable gap

14.7% of queries have a relevant example in the top-100 but not in top-2 — recoverable with a task-specific trained reranker. To reach 90%+ Hit@2, a bi-encoder fine-tuned with contrastive loss on EHRSQL same-template pairs is needed.

4. Entropy Filtering Analysis

Entropy filtering uses the token-level generation entropy as a confidence proxy: high-entropy outputs are abstained rather than returned as SQL. Calibrated on the validation set (1,163 questions) and applied to the test set.

Model	Threshold	EX	RS(10)	Halluc.	Abstentions	Split
ORPO v4 (no filter)	none	40.5%	0.128	42	191/233	Test
ORPO v4 + entropy filter	1.1462	35.6%	0.409	8	225/233	Test
ORPO v4 valid (no filter)	none	39.4%	0.127	41	191/232	Valid
ORPO v4 valid + entropy filter	1.1462	35.6%	0.409	8	225/232	Valid

Entropy filtering improved RS(10) from 0.128 to 0.409 on the EHRSQL 2024 test set — a +28.1 pp gain — but at the cost of EX (40.5% → 35.6%) due to over-abstention. This is the same technique used by the competition winner. The OmniSQL-7B SFT adapter did not

require entropy filtering: it achieved $RS(10)=0.873$ from confident greedy decoding alone.

5. Training Configuration

Parameter	SFT	ORPO
Base model	seeklhy/OmniSQL-7B	OmniSQL-7B + SFT adapter
LoRA rank (r)	32	32
LoRA alpha	64	64
LoRA dropout	0.1	0.1
Target modules	q,k,v,o,gate,up,down (7)	same
Learning rate	1e-4	2e-6
LR schedule	cosine	cosine
Epochs	2	1
Effective batch size	16 (2×8 grad accum)	16
Max sequence length	1,536 tokens	1,536 tokens
Optimizer	paged_adamw_8bit	paged_adamw_8bit
Training examples	53,006 (40,961 ans. + 12,045 unans.)	492 pairs (OmniSQL own wrong preds)
Unanswerable oversampling	3×	N/A
Hardware	Colab A100 (~5 hrs)	Colab A100 (~2 hrs)

6. Key Findings

OmniSQL pre-training is the decisive factor

OmniSQL-7B starts at EX=61.2% zero-shot on the EHRSQL 2024 MIMIC-IV Demo test set — before any fine-tuning. The 2.5M-example SynSQL pre-training gives the model a SQL syntax foundation that makes SFT highly effective. By contrast, Qwen2.5-Coder-7B ORPO v4 only reaches EX=40.5% on the same split despite targeted ORPO training.

SFT-only beats self-training + API models

The competition winner used GPT-3.5-turbo with multi-stage self-training. Our SFT-only pipeline with a 7B open model exceeds their RS(10) by +6 pp. Self-training is not required when the base model already has strong SQL priors.

Abstention accuracy is the RS(10) lever

With N=10, each hallucination costs 10 points. The critical improvement: 133 hallucinations (OmniSQL baseline) → 7 (SFT final). The 3× unanswerable oversampling in SFT data is what drives this.

Entropy filtering has diminishing returns at high abstention quality

Entropy filtering improved ORPO v4 RS(10) from 0.128 → 0.409 (+28 pp) by reducing hallucinations 8 → 42. But OmniSQL-7B SFT already produces only 7 hallucinations without filtering — adding entropy filtering to a well-calibrated model reduces EX without meaningful RS gain.

Cross-encoder rerankers degrade few-shot retrieval

All tested rerankers (MS-MARCO MiniLM, BGE cross-encoder) trained on web data degraded Hit@2 by 2–4 pp vs the hybrid baseline. Domain mismatch is the primary cause. A task-specific bi-encoder trained on EHRSQL pairs is the path to 90%+ retrieval recall.

System prompt fidelity is non-negotiable

Early Colab runs accidentally used a 548-char fallback prompt (no schema) instead of the 2,733-char full MIMIC-IV schema prompt. RS(10) dropped from 0.873 → ~0.21. The full schema context is essential for both SQL accuracy and abstention calibration.

7. What's Next

Next Step	Description	RS(10) Potential
Task-specific retrieval reranker	Fine-tune a bi-encoder (e.g. BGE-M3) with contrastive loss on EHRSQL same-template pairs. Target: Hit@2 $\geq$ 85% vs current 72.1%. Unlocks 14.7% recoverable gap.	+3–5 pp
Self-training pass	Apply SFT-OmniSQL adapter to train_aug (38K questions). Collect high-confidence abstentions as pseudo-labels. Re-train 1 epoch to address train/test distribution gap (8.9% $\rightarrow$ 20% unanswerable ratio). Replicates KAIST winner Stage 2.	+2–4 pp
OmniSQL-32B backbone	Replace 7B with 32B parameter model — same LoRA recipe, requires A100 40GB. OmniSQL-32B approaches GPT-4-level EX on EHRSQL benchmarks per the paper.	+3–6 pp
Production serving	Integrate SFT adapter into the governed copilot pipeline (AST guardrails, k=11 suppression, vLLM AWQ-4bit serving). End-to-end latency target: <2s p50.	N/A

Project: Governed Clinical Analytics Copilot · GitHub: github.com/pbiyyani09/Governed-Clinical-Analytics-Copilot · SOTA analysis: [docs/sota_analysis_ehysql2024.md](#) · July 2026